

Design of a Vending Machine using Finite Automata

Theory of Computing Project Report

Student Name: _____

Register Number: _____

Course: B.Tech CSE

Academic Year: 2025–26

1. Introduction

Finite Automata is a fundamental concept in Theory of Computing used to model systems with finite states. A vending machine operates on inputs such as coins and produces outputs such as products. This project demonstrates how DFA can represent such systems.

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2. Problem Statement

The vending machine accepts coins and dispenses a product when the required amount is reached. The system must behave deterministically and reset after completion.

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3. Theoretical Background

A DFA is defined as a 5-tuple consisting of states, input alphabet, transition function, start state, and final states.

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4. Methodology

The system is modeled by defining states representing money and transitions based on coin input.

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5. Model Design

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6. Implementation

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7. Analysis

The system correctly processes valid inputs and rejects invalid sequences.

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8. Applications

Used in vending machines, ATMs, and embedded systems.

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9. Conclusion

The DFA model effectively represents vending machine behavior.

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10. Future Scope

Can be extended to multi-product and smart systems.

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11. References

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